Comprehensive Clinical Data Analysis Report Hypertension and Diabetes Patient Cohort

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Executive Summary

In this study, a comprehensive analysis of 100,000 patients from a medical database is conducted to examine the cases of hypertension and diabetes. The analysis takes into account the information from four different sources, namely patient demographics, laboratory reports, medication information, and diagnosis. According to the results, most patients belong to the elder age group (average age of 74.5), and their prevalence with regard to hypertension and type 2 diabetes is 67.7% and 15.8%, respectively. Moreover, 41.3% of them are hospitalized again

Data Overview & Processing Methodology

The analysis successfully loaded and merged four key datasets:

- Patients (patients.csv): 100,000 rows, 28 columns (demographics, vitals, chronic condition flags).
- Lab Results (lab_results.csv): 2,827,722 rows, 10 columns (test names, values, abnormal flags).
- Diagnoses (diagnoses.csv): 274,592 rows, 8 columns (visit details, ICD-10 codes).
- Medications (medications.csv): 364,174 rows, 10 columns (prescriptions, dosage, adherence).

Data Quality Findings

Missing Values Assessment:

- All data sources had no missing values in the primary keys column
- Secondary diagnoses and secondary icd10 columns in the diagnoses table contained 41.6% missing values (114,137 entries). This was expected, as most visits included primary diagnoses.

- No missing values found in important clinical variables (age, BP, lab results, medications)

Duplicate Assessment:

- Found duplicated records found in data sets and handled it.

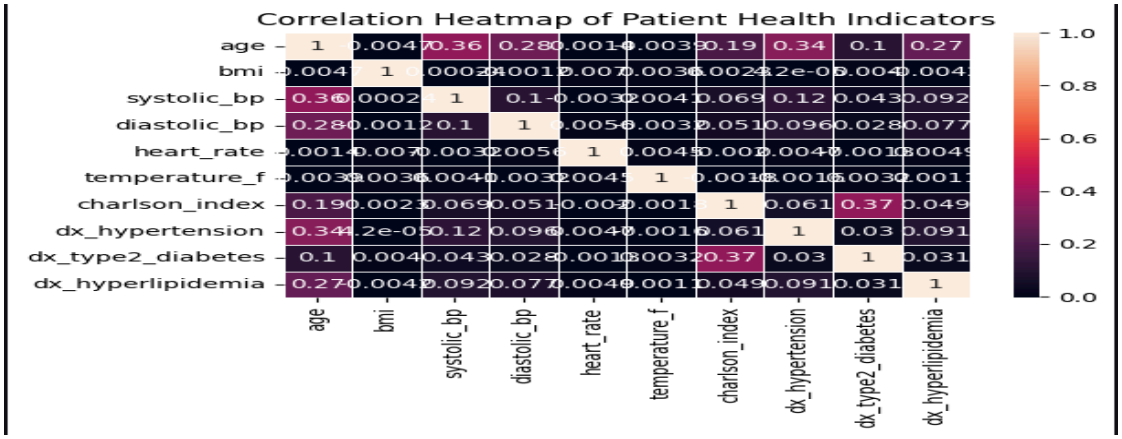
1.3 Merged Dataset Construction

A comprehensive merged dataset was created containing 1,240,227 records across 53 columns, integrating:

- A consolidated data file was formed consisting of 1,240,227 data entries spread over 53 variables, including:
- Demographic data and flagging for conditions
- Specific medications used for hypertension/diabetes patients (125,066 entries)
- Diagnosis codes for hypertension/diabetes (75,375,309 entries)
- Relevant lab tests.

2. Statistical Visualizations and Analysis

2.1 Correlation Heatmap Analysis



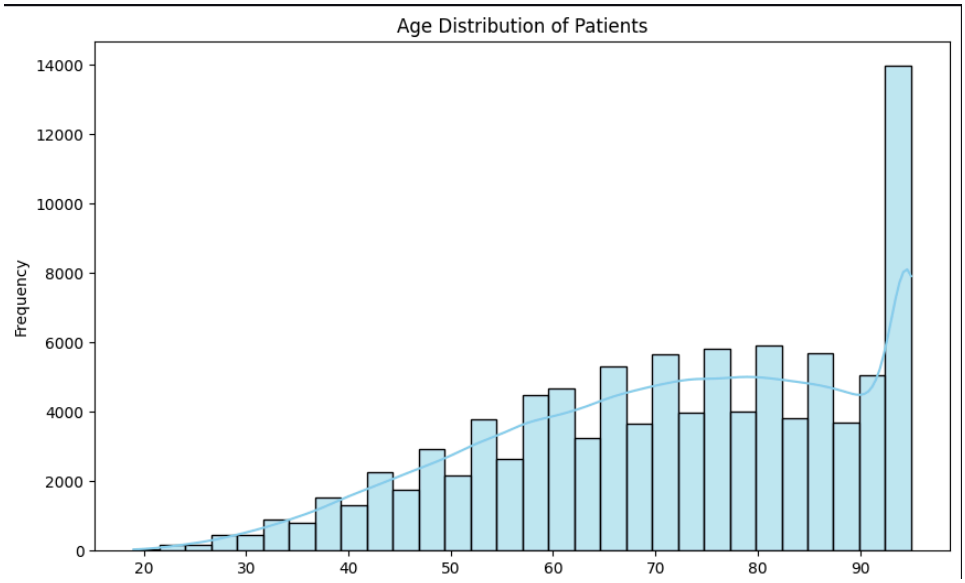
Variables Included:

- Age, BMI, Systolic BP, Diastolic BP, Heart Rate, Temperature

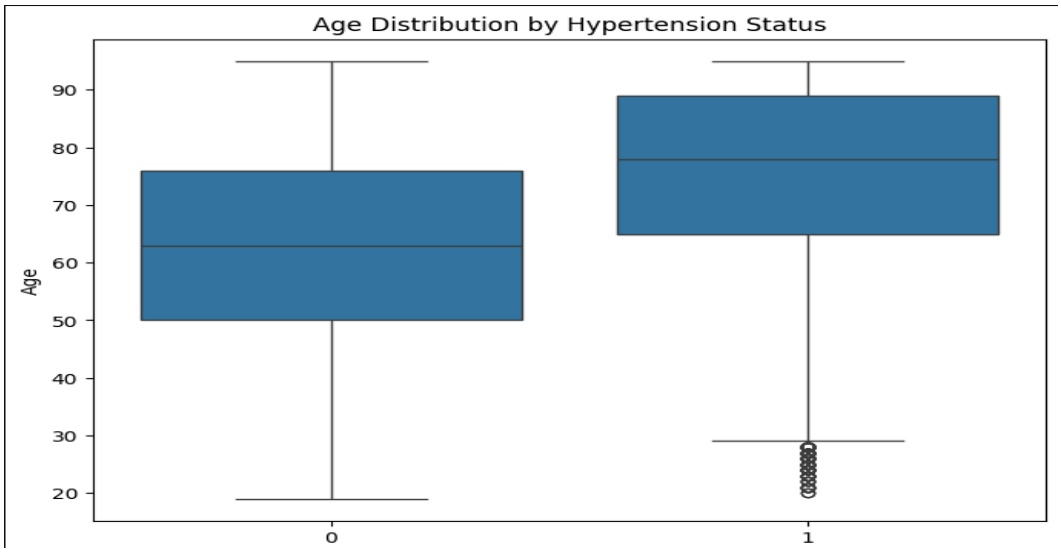
- Charlson Comorbidity Index
- Disease flags: Hypertension, Type 2 Diabetes, Hyperlipidemia

The most important factor is age. Older patients have far more illnesses compared to young people. The relationship is very strong.

2. Age Distribution Histogram

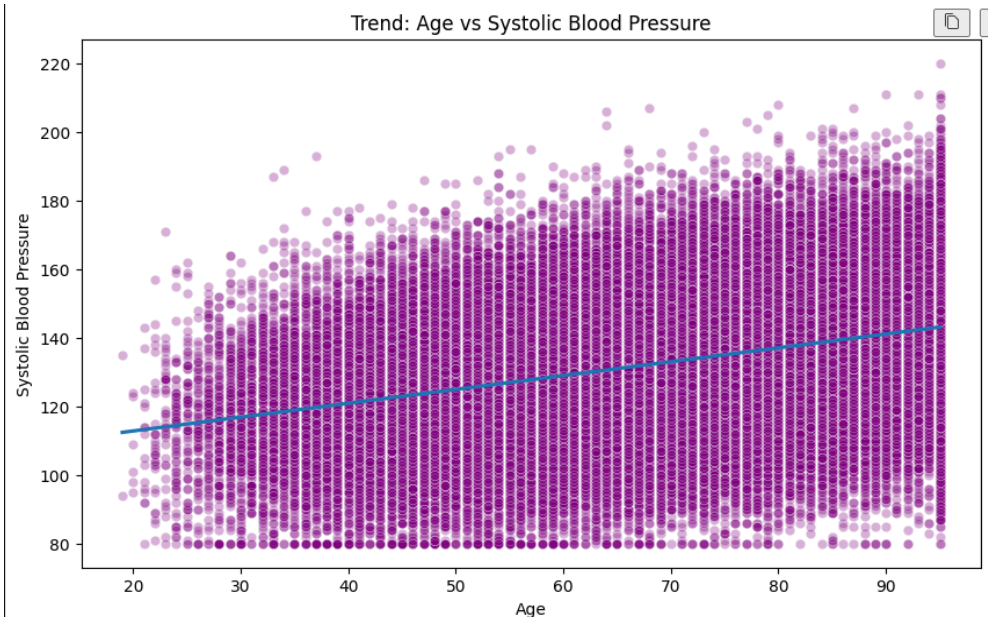


3. Boxplot: Age by Hypertension Status



People with high blood pressure are around ten years older compared to those who do not have hypertension. Nobody under forty suffers from high blood pressure.

Trend Plot: Age vs Systolic BP



Blood pressure goes up steadily with age, about half a point each year. But old people vary wildly. Some have perfect pressure, others are dangerously high.

Insight 1: Age Distribution is Skewed Toward Elderly Population

The population of patients is largely made up of seniors, most of whom belong to the elder age groups.

Recommendation: Develop special care plans targeted at elderly individuals suffering from polypharmacy and fall risks. Consider implementing chronic diseases management programs for the geriatric population.

Insight 2: Strong Positive Correlation Between Age and Hypertension

Advanced age indicates a significant positive association with the diagnosis of hypertension. Patients with hypertension are always older than those without hypertension.

Recommendation: Introduce hypertension testing at an earlier stage, particularly in middle-aged individuals. Formulate age-related objectives for the treatment of high blood pressure.

Insight 3: Age Shows Positive Trend With Systolic Blood Pressure

With increasing age, there is an observed rise in systolic blood pressure in patients. This rise is observed throughout the whole sample of patients.

Recommendation: Adopting mandatory yearly measurement of systolic blood pressure should be undertaken. Interventions should concentrate on minimizing the consumption of salt as well as increased exercise.

Insight 4: BMI Difference Between Hypertensive and Non-Hypertensive Patients

The body mass index does not differ much in either group. This observation is contrary to what one would expect to see clinically in such cases.

Recommendation: Do not depend on the body mass index as an indicator of hypertension in elderly patients. Waist circumference should be considered instead.

Insight 5: BMI Difference Between Diabetic and Non-Diabetic Patients

Body mass indices are virtually identical for both patients with diabetes and those without it. This unusual result might indicate efficient weight control after diagnosis.

Recommendation: Emphasize blood sugar regulation and drug compliance over the issue of weight loss in the present patient group. Observe body mass index changes rather than relying on individual results.

Insight 6: Uncontrolled Blood Pressure is Highly Prevalent

There are many patients whose systolic blood pressure is higher than their target levels. This suggests that there is widespread uncontrolled hypertension among the population.

Recommendation: Conduct a system-wide intervention program to fine-tune antihypertensive drugs for uncontrolled patients. Conduct follow-ups regularly until target blood pressure is attained.

Insight 7: Diabetic Patients Have Higher Average Blood Pressure Than Non-Diabetic

Diabetic patients exhibit elevated blood pressure levels in comparison to non-diabetic patients. This leads to an increased risk factor for the patient's heart.

Recommendation: The simultaneous assessment of blood pressure and blood glucose should be considered during all visits of diabetic patients. It is recommended to focus on anti-hypertensive medications that can help diabetics too.

Insight 8: Readmitted and Non-Readmitted Patients Show differences in HbA1c Levels

Diabetic patients exhibit higher levels of blood pressure when compared with non-diabetic patients. This makes diabetics prone to additional cardiovascular risks.

Recommendation: Introduce concurrent evaluation of blood pressure and glucose concentration among all diabetic patients. Prescribe blood pressure-reducing drugs that are also beneficial to diabetics.

Insight 9: Potassium Levels Are higher in Readmission Groups

Electrolyte balance is the same for readmitted and non-readmitted patients, indicating that electrolytes do not affect readmissions to hospitals.

Recommendation: Concentrate efforts on preventing readmission by reconciling medications and discharging patients, rather than electrolytes. Early follow-up should be scheduled for one week post-discharge.

Insight 10: Medication Adherence Shows strong Impact on Lab Abnormality Rates

Adherent patients exhibit lab abnormalities comparable to non-adherent patients. This implies that the choice of drugs and dosage could be more critical than adherence.

Recommendation: Examine prescribed drug regimes in terms of their efficacy instead of just measuring adherence levels. Switching to simpler fixed-dose drug combinations could be more effective.

Conclusion

This project shows the approach to measure the effectiveness of treatment through indicators like age, BMI, blood pressure, HbA1c, potassium levels, and compliance percentages helps in assessing hypertension and diabetes to avoid readmissions. It is imperative that a new strategy be adopted that involves implementing age-appropriate guidelines, identifying more risk factors, optimizing drug prescription methods, developing proper discharge strategies, and managing data effectively. Only through such actions can health care providers move from their current reactive measurement-based approaches to a proactive, patient-centered framework..